Turbo Tennis

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TURBO TENNIS
EASY | CHALLENGE
PLAYABLE ON WINDOWS & MAC
PLAY NOW!

Game Introduction

Genre: Sports arcade style game (2D tennis) with casual competitive feel.

Game Mechanic:

- The player uses simple controls (e.g., move, swing) to play tennis rallies.
- Two levels: *Easy* (slower ball, forgiving timing) and *Hard* (faster ball, tighter timing).
- Objective is to win X points / first to a defined score.

Scenario/Storyline:
The player arrives at an outdoor dome-stadium tennis facility (grass dome) and competes in either a friendly match (Easy) or a competitive match (Hard). There is no narrative; focus is on gameplay and environment.

Characters:

- Player character: generic tennis player (avatar)
- Opponent: either AI easy or AI hard depending on level

Game Introduction

Goal of the game:

Primary goal

For the player to successfully win a tennis match by reaching the required number of points in either the Easy or Hard level

Secondary goals / motivations

- Improve player skill (especially in Hard level)
- Provide a fun and visually engaging environment
- Encourage replayability by having two difficulty levels

Initial Game Design

Control Interface: Keyboard up, down, right, left arrow keys and spacekey

Graphical Interface:

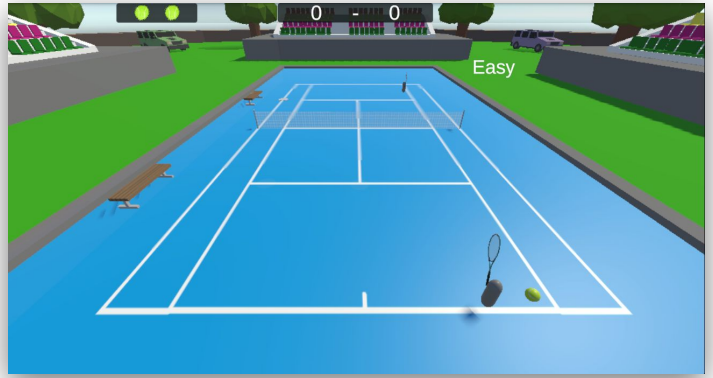
Maps / Levels:

- Map/level is the single tennis court inside the grass dome with bleachers, cars and trees as background/environment.
- Two levels primarily differ by difficulty rather than drastically different map layouts.

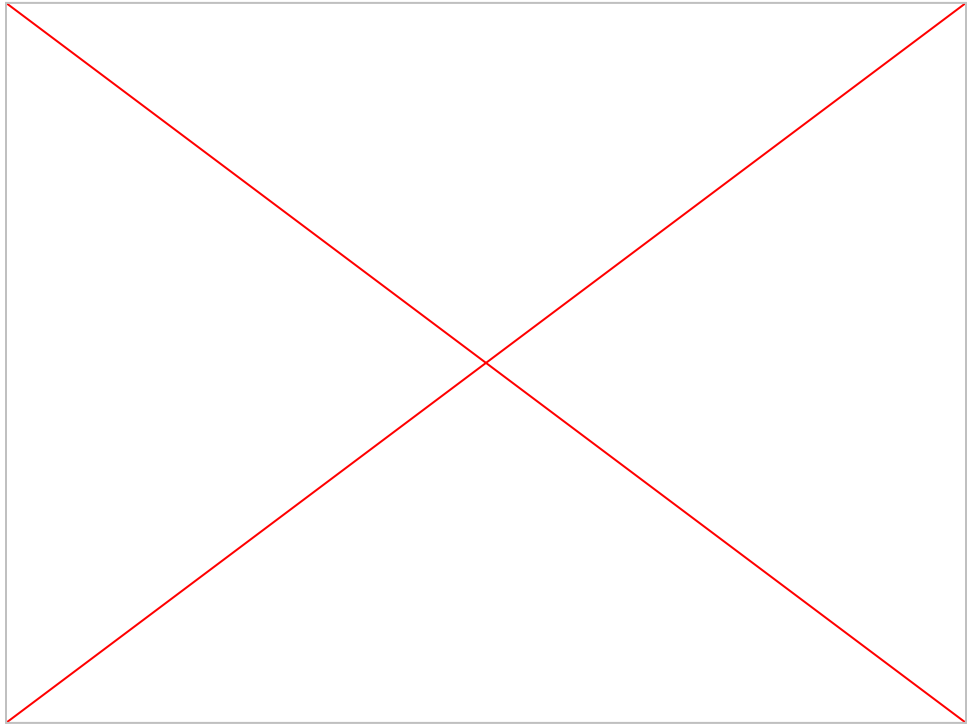
Environment Features:

- Bleachers on four sides give audience ambiance.
- Cars and trees around the dome add visual interest and context.

Initial Game Design



Mid-Demo



Evaluation Plan (User Study Design)

Research Questions:

- Does the difficulty level (Easy vs Hard) affect player enjoyment and perceived competence?
- Does the game environment (bleachers, cars, trees, dome) influence player immersion and satisfaction?
- Is the HUD layout intuitive and does it impact the player's performance and comfort?

Hypotheses:

- H1: Players in the Hard level will report lower perceived competence but higher challenge and possibly higher enjoyment than players in the Easy level.
- H2: A richer environment (with bleachers, cars, trees, dome) will lead to higher immersion and satisfaction than a minimal environment— or simply we expect high environment satisfaction.
- H3: A well-designed HUD will correlate with better player performance (fewer errors, faster reaction) and higher usability/satisfaction ratings.

Evaluation Plan (User Study Design)

Independent variables:

- Difficulty level (Easy vs Hard) → 2 levels

Dependent variables:

- Player performance (e.g., number of points won, number of errors)
- Player enjoyment / satisfaction (survey rating)
- Perceived competence / challenge (survey rating)
- HUD usability (survey rating)

Design Type:

Since each participant could play both levels (Easy and Hard) in randomized order, this would be a **within-subjects design** for difficulty. If you compare different groups (one plays Easy only, another Hard only) then it's between-subjects.

Initial Game Feedback: The Positives

Audio & Visual: The color choices made the environment feel lively and dynamic and matches the tennis ball color; the score indicator is clearly displayed and easy to track, contributing to a smooth and enjoyable overall experience; buttons have a beautiful glossy effect to them; showing the in-game score at the top was found to be very helpful for showing the state of a tennis match and is early visible and does not blend in with the background.

Controls / gameplay: The physics of the ball feel natural and easy to grasp; Both the controls and scoring system are intuitive, allowing players to jump right in without confusion; The controls were smooth and easy to get used to and I liked how the game play flowed

Learnability & feedback: Main menu was easy to navigate and start the game

Initial Game Feedback: Suggested Improvements

Audio & Visual: Font style, size, or contrast were considered hard to read, suggesting we should adjust for better legibility; the HUD scoreboard blends in with the gray wall; labels were suggested so that players can easily tell whose score is whose; background music was suggested to make game more engaging and energetic.

Game flow / functionality problems: the bot (computer opponent) was glitchy sometimes not moving or score not updating properly; after scoring, the game did not always reset for the next serve. If the computer won, the match sometimes continued instead of resetting or signaling end; no obvious way to exit, restart, or return to main menu.

Controls / gameplay issues: Confusion on what controls to use due to lack of tutorial; testers suggested adding control for shot power. Controls and “reset” functionality (after a point or match) were noted as not smooth; refinements are necessary to ensure more responsive controls and seamless gameplay.

Learnability & feedback: There were no instructions before gameplay, making it hard for first-time players to understand controls or flow; lacking feedback on why the ball disappeared.

Revision Plan

The following revisions to Turbo Tennis were based on the feedback from Survey 1:

- **Improve UI readability and clarity:**
    - Adjust font style/size/contrast
    - Ensure scoreboard, buttons, and HUD elements are clearly visible regardless of background
    - Ensure UI scales or lays out properly in both Easy and Hard modes.
- **Enhance controls and game-flow reliability:**
    - Fix control responsiveness
    - Ensure ball-hit detection and reset/serve flow work reliably
    - Ensure correct opponent (bot) movement and scoring logic
    - Implement proper match-end detection and reset or return to main menu.

Revised Game Design

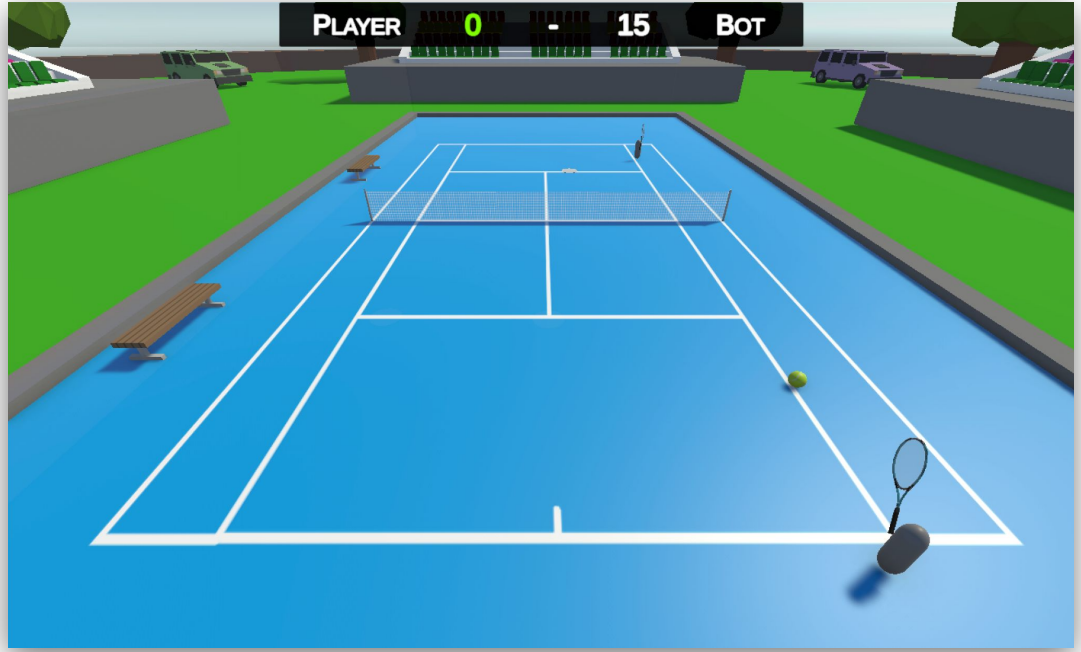
Control interface:

- Control responsiveness refined and swing/hit detection improved to improve swing and miss detection reliability.
- Game reset triggered when point is scored.
- Bot movement and scoring logic fixed to improve consistency in computer opponent behavior and update scores correctly.
- Match-end detection was implemented

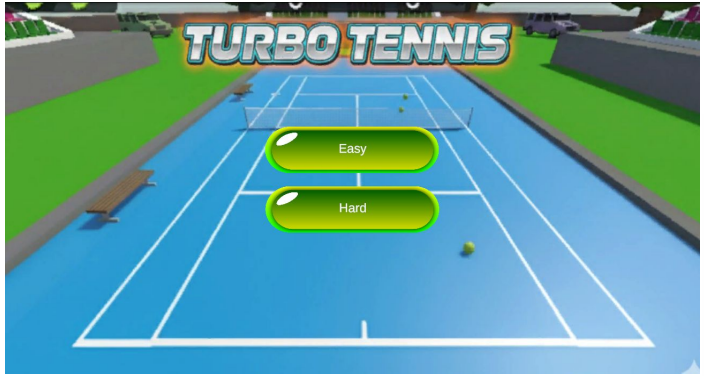
Graphical interface:

- UI / HUD font style, color contrast, and size were adjusted to ensure readability and consistency.
- Added Win / Lose scene transitions and user guidance for replay or exit.
- Differentiated player vs bot score display on the HUD to improve readability and minimize confusion.

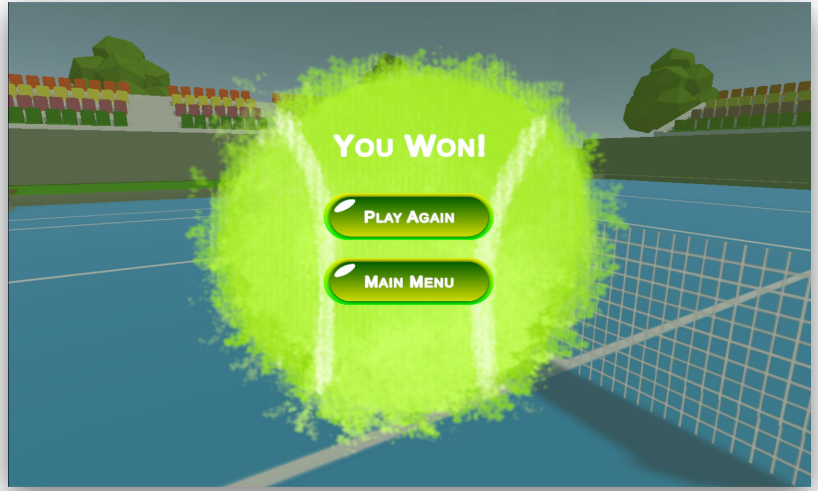
Revised Game Design

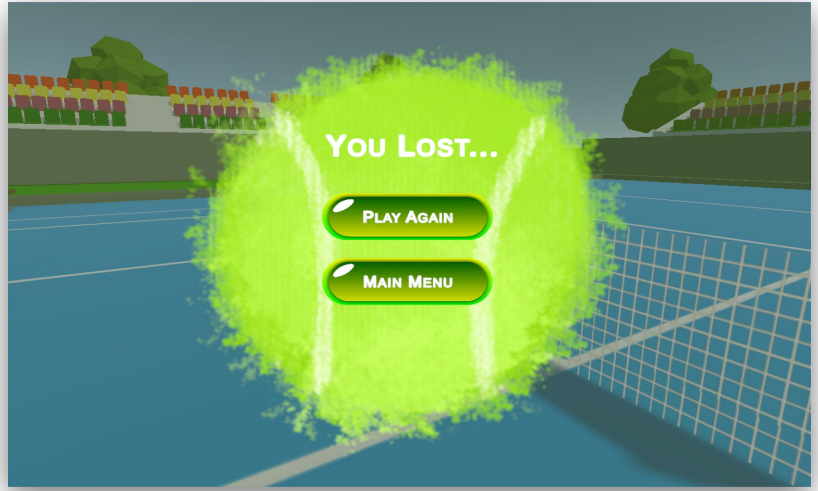


Revised Game Design

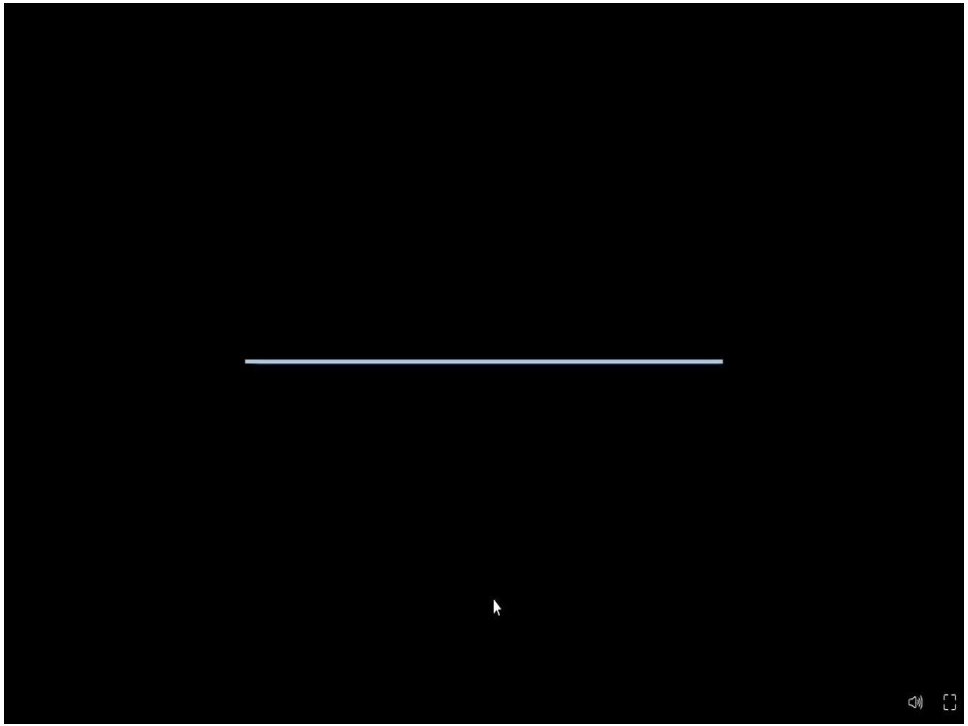


Revised Game Design





Revised Demo



Game Analysis Result

Survey 1

- Difficult to read, testers commented on poor font/contrast
- Swing felt unreliable; sometimes input didn’t register
- Glitches: bot stuck / score not updating / match not resetting properly
- No instructions, “why the ball disappeared” confusion, unclear flow
- Mixed enjoyment; some frustration with controls / flow

Survey 2

- Most testers reported UI was clear and readable
- Most testers felt controls were responsive, swing detection improved
- Game flow stable: serves reset properly, match ends, bot & scoring logic works correctly
- Instructions added + clearer UI feedback; testers felt gameplay easier to understand
- Majority expressed enjoyment and willingness to replay / try different difficulty

Game Analysis Result

Survey 1 : Descriptive Statistic

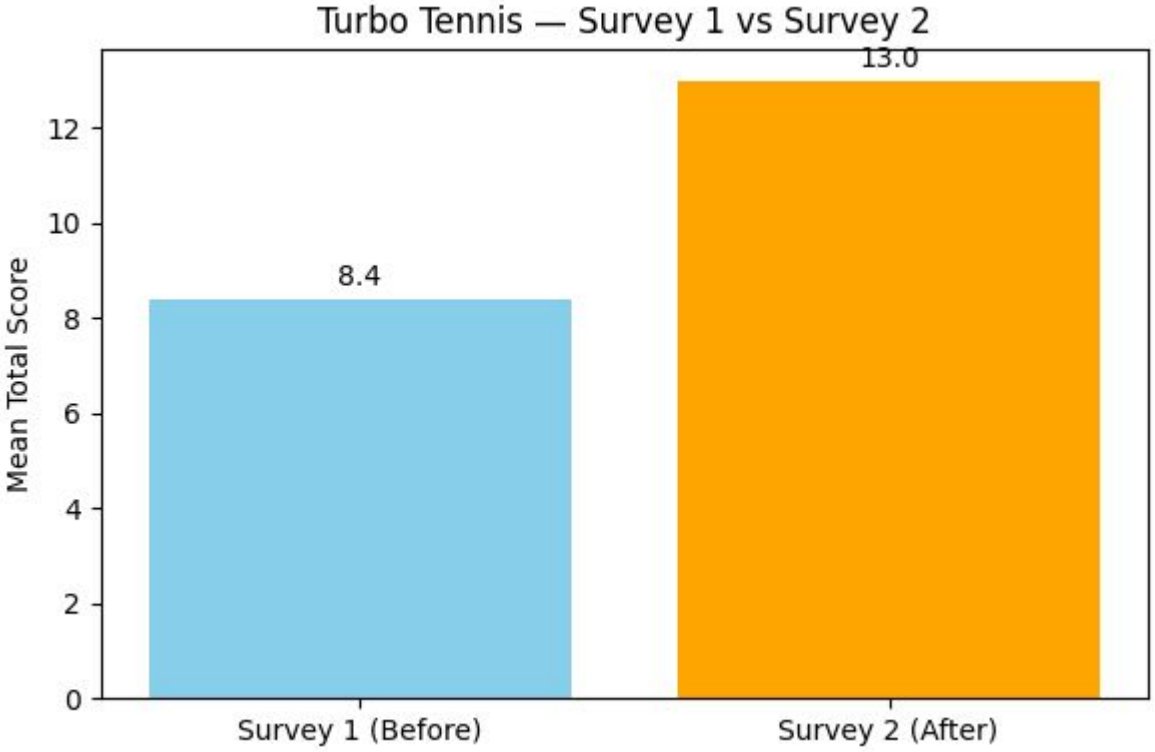
Controls Responsiveness (rate 1 to 5)	2.8 Mean
Easy to Learn (rate 1 to 5)	2.5 Mean
Overall Enjoyment Rate 1 to 5)	3.1 Mean

Survey 2: Descriptive Statistic

Controls Responsiveness (rate 1 to 5)	4.2 Mean
Easy to Learn (rate 1 to 5)	4.5 Mean
Overall Enjoyment Rate 1 to 5)	4.3 Mean

Inferential Statistics: We get the p-value and (optionally) 95% confidence interval for the mean difference.

Game Analysis Result



Discussion

- Iterative design + user feedback is powerful.
- Usability and polish matter as much as core mechanics.
- Small scope enables manageable refinement.
- The improvements we made based on survey feedback (UI, control tweaks, flow fixes) had clear positive effects on how players perceived and enjoyed the game.

Conclusion

- Turbo Tennis is a simple 2D sports game produced from a minimal core mechanic, evolving through iterations and feedback into a player experience that is cohesive, playable and enjoyable.

- There is still space for improvement in the future (audio, visuals, deeper mechanics), yet as things stand, Turbo Tennis meets most of its aims: accessible and enjoyable matches with varying challenge levels that encourage replaying.